

ME 520: Microwave based Manufacturing Processes

Course Code	ME 520
Course Name	Microwave based Manufacturing Processes
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG
Mutual Exclusion	None
Approval	44th BoA (Nov 2021)

Course Contents

- **Introduction and Fundamentals:** Perspective, Material Interactions, Microwave generators, wave propagation, waveguide modes, microwave applicators, Safety aspects in Microwave Material Processing (6 hours)
- **Science and Modelling of microwave material interaction:** Power absorption model and Maxwell's equations, dielectric properties, microwave penetration and power absorbed, material behavior during microwave interaction, heating mechanisms in microwave processing of materials: non-magnetic materials (conduction loss), magnetic materials (hysteresis loss, eddy current loss and residual loss), microwave absorption in characteristically different materials: insulators, metallic powders and bulk metals, composite materials. (10 hours)
- **Manufacturing Processes for Polymers:** Polymers, Polymer composites, microwave assisted compression moulding (MACM), vacuum assisted resin infusion microwave curing (VARIMC), mechanism of processing, roles of process parameters, case studies, lab demonstration. (10 hours)
- **Manufacturing Processes for Metals and their composites:** Surface Engineering, Physics of Microwave Glazing and Cladding. Concept of skin depth, Role of process parameters, advantages and limitations, case studies, lab demonstration. (8 hours)
- **Manufacturing Processes for Ceramics:** Microwave sintering of ceramics, process parameters in sintering, microwave drilling of ceramics, process parameters in microwave drilling, case studies, lab demonstration. (8 hours)

Text Books

1. Metaxas, AC and, and Roger J. Meredith, **Industrial microwave heating**, No. 4. IET, 1983.
2. Pozar, D.M., **Microwave engineering**, John Wiley & Sons, 2011.

References

1. Dieter, G.E. and David J.B., **Mechanical metallurgy**, Vol. 3. New York: McGraw-hill, 1986.
2. DeGarmo, E.P., J. Temple Black, Ronald A. Kohser, and Barney E. Klamecki, **Materials and process in manufacturing**, Prentice Hall, 1997.
3. Chawla, K.K., **Composite materials: science and engineering**, Springer Science & Business Media, 2012.